

METABOLIC PHASE SEPARATION IN THE REGULATION OF STEROL HOMEOSTASIS

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ABSTRACT

Despite decades of research, the precise mechanism by which cells transform graded cholesterol signals into a binary transcriptional response remains unclear. In my talk, I will share our progress in understanding how sterol phase separation within the endoplasmic reticulum membrane forms the physical basis for this switch. A cooperative lipid phase transition occurs at a specific concentration, selectively recruiting and mechanically halting the sterol transcription factor regulatory complex. This process facilitates productive complex formation while preventing transcription. The concentration needed for phase separation aligns with the typical cholesterol-sensing threshold, and enzymatic regulation allows for bidirectional functional dissociation. These results suggest cholesterol sensing emerges from a membrane phase transition.